

EXPLORER UNIVERSE PRODUCTION SYSTEM

Software Architecture Blueprint v1.0

STATUS:	VERSION:	ROLE:	CLASSIFICATION:
APPROVED MASTER	1.0 (Locked)	Chief Software Architect	Studio Operating System

PART 1: SYSTEM OVERVIEW

1.1 Purpose

The **Explorer Universe Production System (EUPS)** is a specialized, enterprise-grade studio operating system designed to govern, architect, validate, and execute the creation of a persistent living virtual universe. It serves as the single source of truth (SSOT) across all studio operations, narrative creation, asset engineering, and production pipelines.

1.2 Vision

EUPS provides an integrated, constitutionally governed environment where creative exploration is backed by mathematical, physical, and narrative discipline. The system operationalizes institutional governance documents ([SC-001](#) through [SC-004](#) , [DC-001](#) , [DC-003](#) , [SB-001](#) through [SB-004](#) , and [G-OPS-001](#)), ensuring that every asset, character, set piece, and story thread is traceably anchored to locked canonical laws.

CORE OPERATING PHILOSOPHY (THE 6 INVIOABLE AXIOMS)

- Authority Before Generation:** No asset or story element may be created without referencing higher authority documents.
- Evaluation Before Approval:** Every candidate must pass automated canon checks and human peer review.
- Approval Before Production:** Unapproved assets cannot enter production build pipelines.
- Locked Authority is Immutable:** Once marked [LOCKED_CANON](#) , core records cannot be overwritten.
- AI Assists; Humans Decide:** AI generates candidates and checks logic; humans hold exclusive write-authority.
- Purpose Before Appearance:** Assets are modeled based on physical, historical, and environmental logic.

1.3 Architectural Differentiation

FUNCTIONAL DIMENSION	TRADITIONAL WIKI / CMS	EXPLORER UNIVERSE PRODUCTION SYSTEM (EUPS)
Data Topology	Flat text linked by hyper-text URLs.	Multi-dimensional Directed Acyclic Graph (DAG) with Asset Dependency Matrices (ADM).
Governance Engine	Collaborative overwrites; basic locks.	10-Tier Authority Hierarchy with hard cryptographic locks and override constraints.
Asset Pipeline	Basic media file attachments.	Master Asset Library (MAL) integration linking geometry, acoustics, and materials.
Lifecycle Control	Draft → Published.	6-Phase State Machine (Idea → Problem → Discovery → Validation → Canon → Production).
AI Integration	Unstructured chatbots/summarizers.	Multi-Agent Subsystems bounded by RAG vector stores and Canon Guardrail Enforcers.

EUPS is architected as an event-driven, microservices-oriented studio engine operating over a unified spatial-temporal graph database.



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MODULE	SCOPE & PURPOSE	PRIMARY DEPENDENCIES	KEY OUTPUT MANIFEST
1. Universe Core	Central registry for universal physical laws and constants.	None (Root Layer)	Universal State Manifest (USM)
2. World Management	Governs macro geographic, climatic, and ecological systems.	Module 1	World Spatial Index (WSI)
3. Civilization Authority	Governs societal, guild house, and institutional laws (SB-001).	Modules 1, 2	Civilization Registry
4. Character Authority	Manages character identities, guild ranks, and biological profiles.	Modules 3, 9	Canonical Character Dossiers
5. Location Authority	Governs micro environments, architectural sets, and acoustics.	Modules 2, 6	Location Architectural Records
6. Asset Authority	Operates SODI asset index, ADM dependencies, and MAL specs.	Modules 1, 5	SODI Index & ADM Graph
7. Story Authority	Manages narrative framing, expedition logs, and visitor lore (SB-004).	Modules 3, 4, 9	Archival Narrative Records
8. Production Mgmt.	Controls candidate-to-canon decision gates (SF-002).	Modules 6, 10	Production Gate Manifests
9. Timeline Mgmt.	Governs spatial-temporal continuity and causal event graphs.	Modules 2, 4	Master Timeline Graph
10. Canon Mgmt.	Issues master cryptographic locks (LOCKED_CANON) and version trees.	All Modules (1–9)	Master State SHA-256 Hash

PART 4: DATABASE ARCHITECTURE

EUPS implements a hybrid persistence model: PostgreSQL for ACID transactional metadata, Neo4j for relational dependency trees, Qdrant for semantic vector search, and S3 for binary assets.

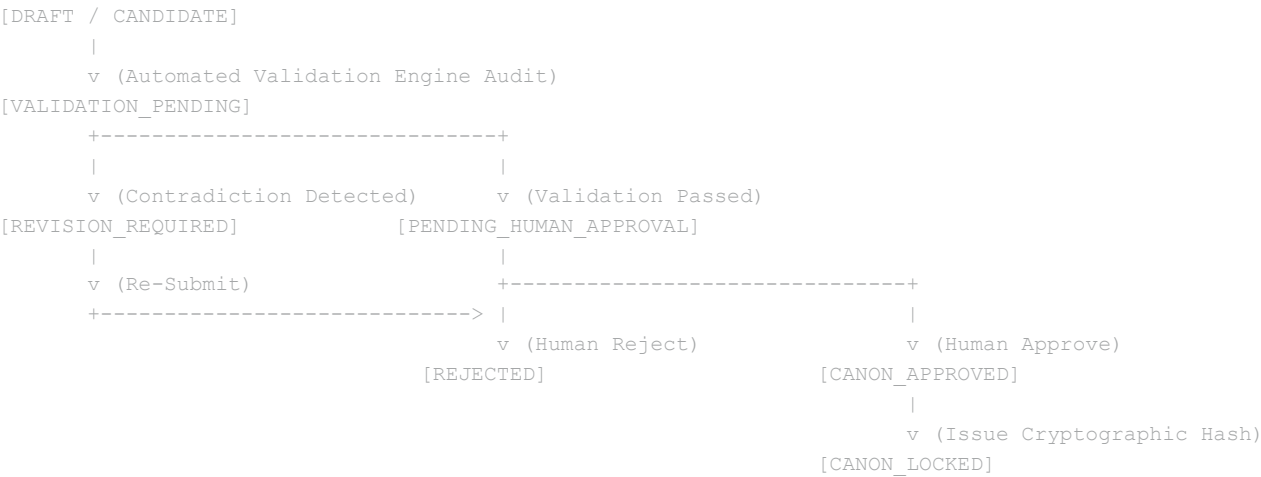
4.1 SQL Entity Schema Specifications

```
-- Entity 1: Studio Operational Database Index (SODI Asset Record)
CREATE TABLE eups_asset (
  asset_id VARCHAR(64) PRIMARY KEY, -- e.g., 'ARCH-010001'
  asset_name VARCHAR(255) NOT NULL,
  authority_tier INT NOT NULL CHECK (authority_tier BETWEEN 1 AND 10),
  category VARCHAR(32) NOT NULL, -- 'ARCHITECTURAL', 'ENVIRONMENTAL', 'PROP'
  family_id VARCHAR(64), -- e.g., 'FAM-000001'
  development_stage VARCHAR(32) NOT NULL, -- 'IDEA', 'PROBLEM', 'DISCOVERY', 'VALIDATION', 'CANON', 'PRODUCT'
  approval_status VARCHAR(32) NOT NULL, -- 'DRAFT', 'PENDING_REVIEW', 'APPROVED', 'REJECTED'
  is_locked BOOLEAN DEFAULT FALSE,
  sha256_hash VARCHAR(64) NOT NULL,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
  created_by_user_id VARCHAR(64) NOT NULL,
  locked_by_user_id VARCHAR(64)
);

-- Entity 2: Candidate Lifecycle Tracking Table
CREATE TABLE eups_candidate_record (
  candidate_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  target_module VARCHAR(64) NOT NULL,
  generation_source VARCHAR(32) NOT NULL, -- 'HUMAN_MANUAL', 'AI_GENERATED', 'HYBRID'
  ai_agent_id VARCHAR(64), -- 'CANON_GUARDIAN', 'CREATIVE_ASSISTANT', etc.
  candidate_payload JSONB NOT NULL,
  validation_results JSONB NOT NULL, -- Contradiction report
  current_gate VARCHAR(32) NOT NULL,
  decision_status VARCHAR(32) NOT NULL DEFAULT 'PENDING',
  reviewer_notes TEXT,
  reviewed_by_user_id VARCHAR(64),
  reviewed_at TIMESTAMP WITH TIME ZONE
);
```



4.2 State Machine Transition Engine



PART 5: USER INTERFACE ARCHITECTURE

Workspaces utilize a standardized **Quad-Status Header** ensuring complete operational visibility across all studio screens:

STANDARD WORKSPACE HEADER LAYOUT

HUMAN CONTROLS	AI ASSISTANCE	VALIDATION STATUS	APPROVAL STATUS
[Edit] [Branch] [Lock]	[Scan] [Draft] [Audit]	PASS: 98% INTEGRITY	LOCKED_CANON - TIER 2

PART 6: AI ARCHITECTURE

AI models interact through a strict middleware API layer wrapped in constraint filters and context injectors. AI engines cannot issue write commands directly to canonical storage.

AGENT IDENTIFIER	FUNCTIONAL SCOPE	KNOWLEDGE SOURCE / VECTOR RAG	SYSTEM AUTHORITY SCOPE
Universe Assistant AI	System navigation & user query resolution.	API Schemas & Studio Documentation	Read-Only across unlocked modules.
Canon Guardian AI	Contradiction scanning & non-contamination checks.	Qdrant Vector Store (Books I-IV , SB-001 .. 004)	System Auditor (Writes to Diagnostic Logs).
Research AI	Deep archival lore search & timeline correlation.	Full Graph DB & SQL Canon Stores	Read-Only with cryptographic citation links.
Creative Assistant AI	Generates candidate text, dialogue, & location concepts.	Context-Injected RAG Store	Candidate Writer (Writes strictly to Candidate Queue).
Production Assistant AI	Asset dependency checks & 3D mesh analysis.	DC-003 Production Bible & ADM Graph DB	Production Auditor (Read-Only).

PART 7: SECURITY AND PERMISSION MODEL

Access control combines Role-Based Access Control (RBAC) with Attribute-Based Authority Tiers (ABAC):

PERMISSION LEVEL	CREATOR ROLE	REVIEWER ROLE	APPROVER ROLE	ADMINISTRATOR ROLE
Draft Candidate	ALLOWED	ALLOWED	ALLOWED	ALLOWED
Review Candidate	DENIED	ALLOWED	ALLOWED	ALLOWED
Advance Gate Phase	DENIED	DENIED	ALLOWED	ALLOWED
Lock Canon Record	DENIED	DENIED	ALLOWED (TIER 1-3)	ALLOWED
Override Lock Conflict	DENIED	DENIED	DENIED	ALLOWED (TIER 1-2)

PART 8: DEVELOPMENT ROADMAP

VERSION	TARGET SCOPE	KEY DELIVERABLES
v0.1	Foundation System	Core SQL/Graph DB schemas, RBAC auth, document reader.
v0.2	Authority Engine	SODI index, ADM graph, SHA-256 locking service, authority override logic.
v0.3	Story & Production System	6-Phase Gate State Machine, Story script workspace, Location spec builders.
v0.4	AI Intelligence Layer	Multi-Agent orchestration, Qdrant Vector RAG, Canon Guardian scanner.
v1.0	Complete Production Release	Full enterprise workspace console, USD exporters, production readiness hardening.

SPECIFICATION AUTHOR:
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Digital Signature Verified

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APPROVED FOR EXECUTION